



BACHELOR OF COMPUTER SCIENCE

**SUBMITTED TO
ARIA UNIVERSITY**

SUBMITTED BY

Sebghatullah “Yarzada”

Title: Inventory Management Mobile App

**SUPERVISED BY
Dr. Yama Ramin**

October 2022



ARIA UNIVERSITY, AMIRY CITY, BALKH, AFGHANSTAN



A PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE
BACHELOR DEGREE.

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CERTIFICATE

I hereby certify that the work which is being presented in the BCS Project Report entitled “IMS Guide Mobile Application”, in partial fulfillment of the requirements for the award of the Bachelor of Computer Science and submitted to the Department of BCS, Aria University, Afghanistan is an authentic record of my own work carried out during a period from November 2018 to November 2022(8th semester) under the supervision of (Dr. Yama Ramin), BCS Department.

The matter presented in this Project Report has not been submitted by me for the award of any other degree elsewhere.

Signature of Student's
Sebghatullah “Yarzada”

This is to certify that the above statement made by the student(s) is correct to the best of my knowledge.

Signature of Supervisor

Date: / / 2022

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Sebghatullah “Yarzada”

ABBREVIATIONS

IDE: 'Integrated Development Environment'

API: Application Program Interface

I/O: Input/output

UI: User Interface

UX: User Experience

HTTP: Hypertext Transfer Protocol

URL: Uniform Resource Locator

WORA: Write Once, Run Anywhere

OS: Operating System

ISO: International Standards Organization

VCS: version control system

VCS: version control software

APK: Android package kit

SDK: Software Development Kit

KDE: kernel Density Estimation

JDK: Java Development Kit

JRE: Java Runtime Environment

IOS: iPhone Operating System

SQL: Structured Query Language

ABSTRACT

Inventory Management Mobile Application is an online service , By Using this App any shop or sales manager can sign up to their own account and can manage their own stocks, this Application has the ability to store & retrieve or calculate (debit and credit) all stocks, sales, loans or expenses. And any account holder can access their accounts from multiple devices or from any where. the goal of this master's project is to design an online Stocks application named “**Inventory Management Mobile Application**”,

The project is Developed using IDE Android Studio, Flutter platform with Dart Language and save its data to Firebase Firestore that can be retrieve very fast.

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CHAPTER 1

1.1 System Analysis

For developing and designing Inventory Management Mobile Application”, I have used Android Studio IDE and programming languages Dart. I have used SQLite database storage for storing the information about all data.

1.2 Feasibility Study

Economic Study: Inventory Management”, By using this app anyone has access to all.

Operational Study: Inventory Management debit and credit of IMS directly in their mobile.

Technical Study: Inventory Management does not need professional user to use it. Every person can use it in mobile.

1.3 System Requirement

To use the Mobile Application, you must have the minimum software and Hardware required for that, and if you do not have the minimum required, you cannot use the program. No special conditions are required to use this Mobile Application. To use it, We must have a Mobile Application.

1.4 Software Requirement

Inventory Management is made only for users. To install and use IMS mobile application system all users can install and use it. Users doesn't need many requirements for using this Online mobile application. The Only Requirement is to have access to an internet connection.

1.5 Hardware Requirement

Because this IMS mobile application it needs mobile hardware storage and user cand find all data one time installation.

- OS Android / iPhone
- RAM 4GB
- Storage 80MB

1. 6 Code Description

This chapter describes all programming language and IDE we have used for developing this mobile application and each classis and functionality inside the application.

1.7 Android

Android is an open source and Linux-based **Operating System** for mobile devices such as smartphones and tablet computers. Android was developed by the *Open Handset Alliance*, led by Google, and other companies. Android offers a unified approach to application development for mobile devices which

means developers need to develop only for Android, and their applications should be able to run on different devices powered by Android. The first beta version of the Android Software Development Kit (SDK) was released by Google in 2007, whereas the first commercial version, Android 1.0, was released in September 2008. On June 27, 2012, at the Google I/O conference, Google announced the next Android version, 4.1 **Jelly Bean**. Jelly Bean is an incremental update, with the primary aim of improving the user interface, both in terms of functionality and performance. The source code for Android is available under free and open-source software licenses. Google publishes most of the code under the Apache License version 2.0 and the rest, Linux kernel changes, under the GNU General Public License version 2.



1.8 Android Studio

Step 1 - System Requirements

Android Studio is a Mobile development tool. You will be delighted, to know that you can start your Android application development on either of the following operating systems...

- Microsoft® Windows® 10/8/7/Vista/2003 (32 or 64-bit)
- Mac® OS X® 10.8.5 or higher, up to 10.9 (Mavericks)
- GNOME or KDE desktop

Second point is that all the required tools to develop Android applications are open source and can be downloaded from the Web. Following is the list of

software's you will need before you start your Android application programming.

- Java JDK5 or later version
- Java Runtime Environment (JRE) 6
- Android Studio

Step 2 - Setup Android Studio

Overview

Android Studio is the official IDE for android application development. It works based on IntelliJ IDEA, You can download the latest version of android studio from [Android Studio 2.2 Download](#), If you are new to installing Android Studio on windows, you will find a file, which is named as android-studio-bundle-143.3101438windows.exe. So just download and run on windows machine according to android studio wizard guideline.

If you are installing Android Studio on Mac or Linux, You can download the latest version from [Download](#), or [Android Studio Linux Download](#), check the instructions provided along with the downloaded file for Mac OS and Linux. This tutorial will consider that you are going to setup your environment on Windows machine having Windows 8.1 operating system.

1.9 Installation

So, let's launch Android Studio.exe, Make sure before launch Android Studio, Our Machine should require installed Java JDK. To install Java JDK, take a reference of [Android environment setup](#)

Once you launched Android Studio, it's time to mention JDK7 path or later version in android studio installer. install JDK to android SDK.

- Need to check the components, which are required to create applications, below the image has selected **Android Studio, Android SDK, Android Virtual Machine and performance (Intel chip)**.
- Need to specify the location of local machine path for Android studio and Android SDK, below the image has taken default location of windows 8.1 x64 bit architecture.
- Need to specify the ram space for Android emulator by default it would take 512MB of local machine RAM.
- At final stage, it would extract SDK packages into our local machine, it would take a while time to finish the task and would take 2626MB of Hard disk space.
- After done all above steps perfectly, you must get finish button and it gonna be open android studio project with Welcome to android studio message as shown below

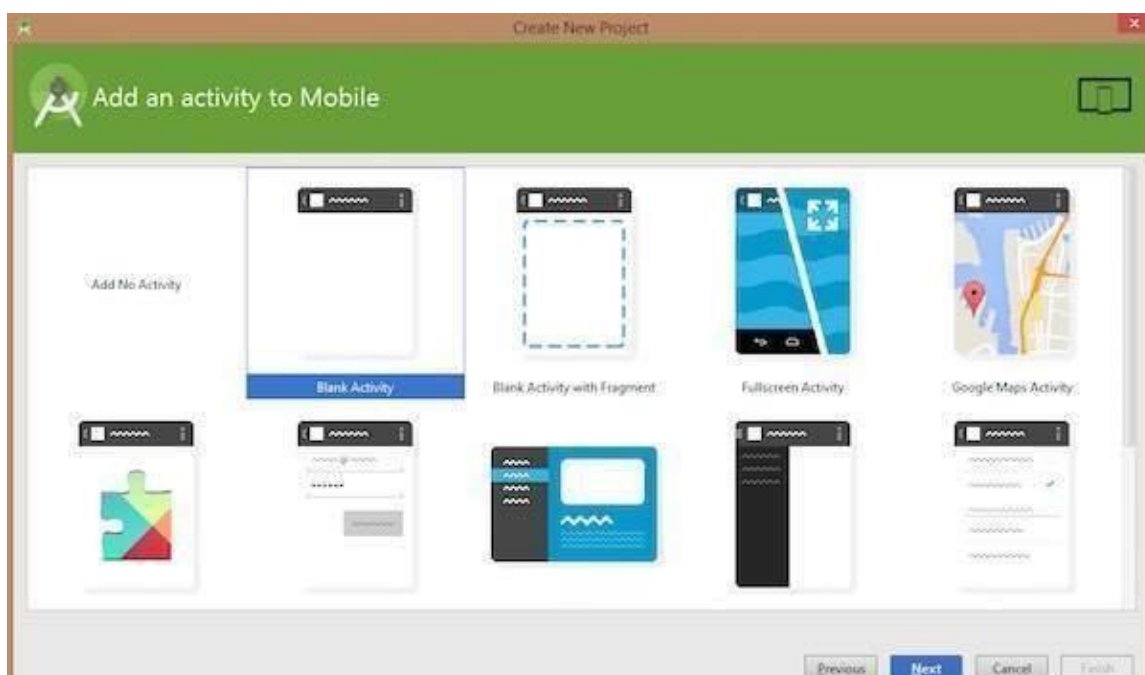
- You can start your application development by calling start a new android studio project. in a new installation frame should ask Application name, package information and location of the project.
- After entered application name, it going to be called select the form factors your application runs on, here need to specify Minimum SDK, in our tutorial, I have declared as API23: Android 6.0(Mashmallow)
- The next level of installation should contain selecting the activity to mobile, it specifies the default layout for Applications
- At the final stage it going to be open development tool to write the application code.

- Step 3 - Create Android Virtual Device

To test your Android applications, you will need a virtual Android device. So before we start writing our code, let us create an Android virtual device. Launch Android AVD Manager Clicking AVD_Manager icon as shown below...

-After Click on a virtual device icon, it going to be shown by default virtual devices which are present on your SDK, or else need to create a virtual device by clicking Create new Virtual device button.

-If your AVD is created successfully it means your environment is ready for Android application development. If you like, you can close this window using top-right cross button. Better you restart your machine and once you are done with this last step, you are ready to proceed for your first Android example but before that we will see few more important concepts related to Android Application Development.



1.10 Flutter Framework

Flutter is an open-source framework to create high quality, high performance mobile applications across mobile operating systems - Android and iOS. It provides a simple, powerful, efficient and easy to understand SDK to write mobile application in Google's own language, Dart.

In general, developing a mobile application is a complex and challenging task. There are many frameworks available to develop a mobile application. Android provides a native framework based on Java language and iOS provides a native framework based on Objective-C/Swift language. However, to develop an application supporting both the OSs, we need to code in two different languages using two different frameworks. To help overcome this complexity, there exists mobile frameworks supporting both OS. These frameworks range from simple HTML based hybrid mobile application framework (which uses HTML for User Interface and JavaScript for application logic) to complex language specific framework (which do the heavy lifting of converting code to native code). Irrespective of their simplicity or complexity, these frameworks always have many disadvantages, one of the main drawbacks being their slow performance. In this scenario, Flutter – a simple and high-performance framework based on Dart language, provides high performance by rendering the UI directly in the operating system's canvas rather than through native framework. Flutter also offers many ready to use widgets (UI) to create a modern application. These widgets are optimized for mobile environment and designing the application using widgets is as simple as designing HTML. To be specific, Flutter application is itself a widget. Flutter widgets also supports animations and gestures. The application logic is based on reactive programming. Widget may optionally have a state. By changing the state of the widget, Flutter will automatically (reactive programming) compare the widget's state (old and new) and render the widget with only the necessary changes instead of re-rendering the whole widget. We shall discuss the complete architecture in the coming chapters.

1.11 Features of Flutter

Flutter framework offers the following features to developers:

- Modern and reactive framework.
- Uses Dart programming language and it is very easy to learn.
- Fast development.
- Beautiful and fluid user interfaces.
- Huge widget catalog.
- Runs same UI for multiple platforms.
- High performance application.

Disadvantages of Flutter

Despite its many advantages, flutter has the following drawbacks in it:

- Since it is coded in Dart language, a developer needs to learn new language (though it is easy to learn).